

Name: _____

Law of Sines. Round to the nearest tenth.

1. $m\angle A = 49^\circ$, $m\angle B = 72^\circ$, $a = 18$. Find $m\angle C$, b , and c .
2. $m\angle B = 35^\circ$, $m\angle C = 46^\circ$, and side a (between them) is 30. Find $m\angle A$ and b .

Law of Cosines. Round to the nearest tenth.

3. $b = 9$, $c = 16$, $m\angle A = 38^\circ$. Find a .
4. $b = 21$, $c = 13$, $m\angle A = 114^\circ$. Find a .
5. Sides 7, 9, 11: find the largest angle.
6. Sides 10, 12, 17: find the angle opposite the 12.

Area and applications.

7. Find the area of a triangle with sides 14 and 20 and a 58° included angle.
8. Two fire stations are 40 mi apart. Station A sights a fire at 63° from the line AB; Station B sights it at 71° . How far is the fire from Station A? (Figure at right.)
9. A triangular field has sides 120 m and 95 m meeting at 74° . Find the third side.
10. For each situation, name the law: (a) ASA (b) SAS (c) SSS (d) AAS.

